

Team Overview

Flight & Airport Big Data Analytics Platform · 48-Hour Sprint

READ THIS FIRST

This document holds the **shared contract**: the tables, the column names and types, and the metric definitions. All 4 members follow it exactly. Your personal brief (files 01–04) tells you what to do. This document shows what everyone else does and what we all agreed.

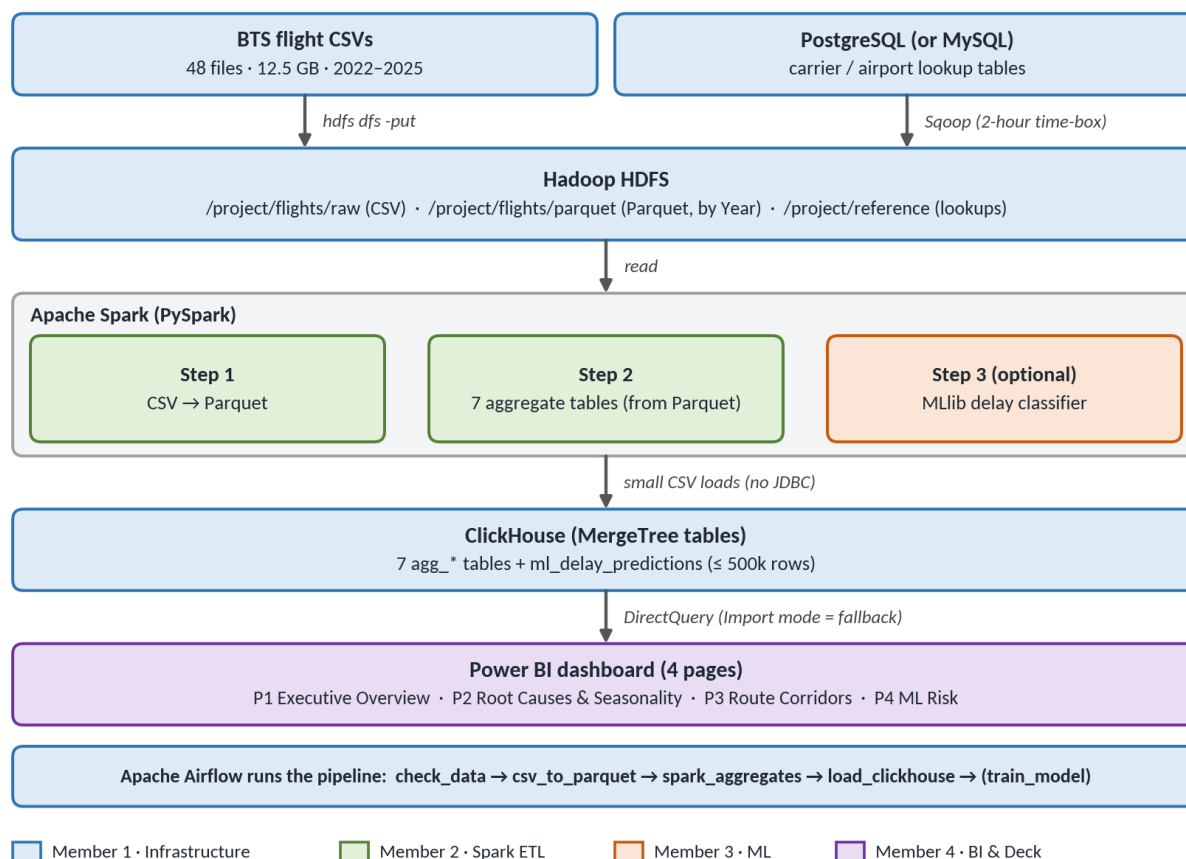
1. Project summary

- **What:** Final Graduation Project, Big Data Training Program, National Telecommunication Institute (NTI), Egypt. Team of 4. **Strict deadline: 48 hours from kickoff.**
- **Goal:** a working Big Data pipeline + a Power BI dashboard + a final deck. The PySpark MLlib model is an optional extension.
- **Why:** delays and cancellations cost airlines and passengers billions. We rank airlines and airports, find delay root causes, show time patterns, cancellations and routes.
- **Data:** U.S. BTS Reporting Carrier On-Time Performance, 2022–2025. **48 monthly CSV files, ~12.5 GB, ~27.4M flights, 109 fields.** Files are in `data/raw/`.
- **Dev sample:** `2024_1.csv` (~550k rows). Day 1: everyone builds on it. Day 2: switch to all 48 months.
- **Batch only.** Kafka / Spark Structured Streaming is not built. It appears only on the future-work slide (slide 12).
- **How we work:** contract first. At **hour 1** all members agree the 8 tables and the metric definitions. Changing the contract later costs four people's work.
- **Time reality:** 48 clock hours is not 48 working hours. Plan sleep in shifts.

2. Who owns what

Member	Owns	Main deliverables
Member 1 Infrastructure & Orchestration 01_Member1_Infrastructure Name: _____	Environment, HDFS, PostgreSQL, Sqoop, ClickHouse setup + DDL, Airflow	• VM health check or Docker stack • HDFS layout, ingestion, lookup tables • ClickHouse DB + 8 tables (DDL) • <code>flight_pipeline_dag.py</code> (BashOperator tasks)
Member 2 Spark Analytics & ETL 02_Member2_Spark_ETL Name: _____	CSV→Parquet, 7 aggregate tables, CSV exports, ClickHouse loads	• <code>spark_csv_to_parquet.py</code> • <code>spark_aggregates.py</code> (Tables 1–7) • Sample CSVs for Member 4 (hour 8 / 12) • Verification queries for Q1–Q8
Member 3 ML Engineer 03_Member3_ML Name: _____	MLlib pipeline (optional extension)	• Leak-free delay classifier (<code>ArrDel15</code>) • AUC-ROC, precision / recall, confusion matrix • Feature importance / coefficients • <code>ml_delay_predictions</code> (≤ 500k rows)
Member 4 BI & Presentation Lead 04_Member4_BI_and_Deck Name: _____	Power BI + final deck + README	• Power BI ↔ ClickHouse test (hours 0–3) • 4-page Power BI dashboard • 12-slide deck + technical README

3. Architecture



Flow: CSV → HDFS → Parquet → Spark → ClickHouse → Power BI. Airflow runs the steps in order. Colours show the owner of each box.

- After the first read, **every job reads Parquet, never raw CSV**. Raw CSV runs take ~30 minutes. Parquet is ~1.5–2 GB.
- Spark → ClickHouse uses **small CSV loads**, not JDBC. JDBC often fails on type mismatches.
- Power BI uses **DirectQuery** on ClickHouse. If it fails: export the tables to CSV and use **Import mode**.

4. Key handoffs

From → to	What	By hour
All members	Agree the 8 tables (names, columns, types) and the metric definitions	Hour 1
Member 1 → Member 4	ClickHouse running, database <code>flight_analytics</code> + a dummy table, port 8123 reachable from Windows. Member 4 proves the Power BI connection.	Hours 0–3 (proven by hour 3)
Member 1 → Member 2	1-month sample in HDFS: <code>/project/flights/sample</code>	Hours 3–8 †
Member 2 → Member 4	Sample CSVs of Tables 1 and 4	Hour 8
Member 2 → Member 4	All sample CSVs	Hour 12
Member 1 → Members 2, 3	All 8 ClickHouse tables created from the DDL	Hours 14–20 †
Member 2 → Member 4	Sample aggregates loaded into ClickHouse (Member 4 tests end to end in hours 20–24)	Hours 14–20
Members 2, 3 → Member 1	Spark scripts and the training script, so the Airflow DAG can call them	Hours 20–24 †
Member 3 → Member 4	Predictions loaded into <code>ml_delay_predictions</code>	Hours 20–24
Member 1 → Members 2, 3	48 CSVs in <code>/project/flights/raw/</code>	Hours 24–30 †
Member 2 → Member 4	Final aggregates loaded into ClickHouse, row counts verified	Hours 30–36
Member 3 → Member 4	Confusion matrix, precision / recall, top drivers; then ROC curve and feature-importance visuals	Hours 30–42 †

From → to	What	By hour
Member 1 → Member 4	Run times and storage sizes; green Airflow tree-view screenshot	Hours 30–42 †
Member 2 → team	Q1–Q8 verification queries, for the report	Hours 36–42
Members 1, 2, 3 → Member 4	Content for the slides each one owns (3–5, 6–8, 9–10)	Before slides 1–6 (hours 24–30) and 7–12 (hours 36–42) †

† Not written as a handoff in the plan, but the schedule needs it. Agree the exact hour with the other member.

5. The 8 key questions and the tables that answer them

#	Question	Answered by
Q1	Which airlines have the highest delay rates and average delay duration?	<code>agg_airline_performance</code>
Q2	Which origin/destination airports experience the most delays?	<code>agg_airport_performance</code>
Q3	What are the peak hours for flight delays?	<code>agg_hourly_delays</code>
Q4	What is the dominant root cause of delays?	<code>agg_delay_causes_monthly</code>
Q5	How do delay probabilities change by month, season, and day of week?	<code>agg_calendar_delays</code> (season derived from Month in Power BI)
Q6	What is the cancellation rate per airline/airport, and the main reasons?	Rates: <code>agg_airline_performance</code> , <code>agg_airport_performance</code> . Reasons: <code>agg_cancellation_reasons</code>
Q7	What are the highest-traffic routes?	<code>agg_route_traffic</code>
Q8	How has domestic traffic volume evolved year over year (2022–2025)?	<code>agg_airline_performance</code> (sum <code>Total_Flights</code> by Year) or <code>agg_calendar_delays</code>

6. Table contract: 7 summary tables + 1 ML table

Column names and types are fixed. Do not rename, add or retype columns. The full DDL is in

01_Member1_Infrastructure.

Table 1 · `agg_airline_performance`

Answers: Q1, Q6 (rates), Q8 (sum `Total_Flights` by Year) · Grain: Year × Airline

Year	UInt16	Reporting_Airline	String	Airline_Name	String
Total_Flights	UInt32	OnTime_Flights	UInt32	Delayed_Flights	UInt32
Delay_Rate_Pct	Float32	Avg_Dep_Delay_Min	Float32	Avg_Arr_Delay_Min	Float32
Cancelled_Flights	UInt32	Cancellation_Rate_Pct	Float32		

Table 2 · `agg_airport_performance`

Answers: Q2, Q6 (rates) · Grain: Airport (all years)

Airport_Code	String	Airport_City	String	Airport_State	String
Total_Departures	UInt32	Total_Arrivals	UInt32	Dep_Delay_Rate_Pct	Float32
Arr_Delay_Rate_Pct	Float32	Avg_Dep_Delay_Min	Float32	Cancelled_Departures	UInt32

Table 3 · `agg_delay_causes_monthly`

Answers: Q4 · Grain: Year × Month

Year	UInt16	Month	UInt8	Total_Delay_Minutes	UInt64
Carrier_Delay_Min	UInt64	Weather_Delay_Min	UInt64	NAS_Delay_Min	UInt64
Security_Delay_Min	UInt64	Late_Aircraft_Delay_Min	UInt64	Carrier_Pct	Float32
Weather_Pct	Float32	NAS_Pct	Float32	Security_Pct	Float32
Late_Aircraft_Pct	Float32				

Table 4 · `agg_hourly_delays`

Answers: Q3 · Grain: Scheduled dep. hour (0–23)

Dep_Hour	UInt8	Total_Flights	UInt32	Delayed_Flights	UInt32
Delay_Probability_Pct	Float32	Avg_Delay_Minutes	Float32		

Table 5 · agg_route_traffic

Answers: Q7 · Grain: Directed route (Origin → Dest)

Origin	String	Dest	String	Route_Name	String
Flight_Count	UInt32	Avg_AirTime_Min	Float32	Distance_Miles	UInt32
Route_Delay_Rate_Pct	Float32				

Table 6 · agg_calendar_delays

Answers: Q5, Q8 · Grain: Year × Month × Day_Of_Week

Year	UInt16	Month	UInt8	Day_Of_Week	UInt8
Total_Flights	UInt32	Delayed_Flights	UInt32	Cancelled_Flights	UInt32
Delay_Rate_Pct	Float32	Avg_Arr_Delay_Min	Float32		

Table 7 · agg_cancellation_reasons

Answers: Q6 (reasons) · Grain: Year × Airline × Cancellation code

Year	UInt16	Reporting_Airline	String	Cancellation_Code	String
Cancellation_Reason	String	Cancelled_Flights	UInt32		

Table 8 · ml_delay_predictions

Answers: Power BI Page 4 (≤ 500k-row sample of the 2024 test set) · Grain: One sampled test flight (≤ 500k rows)

FlightDate	Date	Reporting_Airline	String	Origin	String
Dest	String	Dep_Hour	UInt8	Actual_Delayed	UInt8
Predicted_Delayed	UInt8	Delay_Probability	Float32	Risk_Category	String

7. Metric definitions

Everyone uses exactly these, so every member gets identical numbers.

- **Total_Flights** = all rows, including cancelled and diverted.
- **Operated flight** = **Cancelled** = 0 AND **Diverted** = 0 AND **ArrDel15** not null.
- **Delayed_Flights** = operated with **ArrDel15** = 1; **OnTime_Flights** = operated with **ArrDel15** = 0.
- **Delay_Rate_Pct** = Delayed / (Delayed + OnTime) × 100. **Cancellation_Rate_Pct** = Cancelled / Total_Flights × 100.
- **Dep_Delay_Rate_Pct / Arr_Delay_Rate_Pct** = share with **DepDel15** = 1 / **ArrDel15** = 1.
- **Avg_*_Delay_Min** = average of **DepDelayMinutes** / **ArrDelayMinutes** (early = 0) over operated flights.
- **Total_Delay_Minutes** = sum of the five cause columns (null → 0). **Cause_Pct** = cause minutes / Total_Delay_Minutes × 100.
- **Dep_Hour** = $\text{floor}(\text{CRSDepTime} / 100) \% 24$. **Day_Of_Week**: 1 = Monday ... 7 = Sunday (BTS **DayOfWeek**).
- **Season** (Power BI): Dec–Feb Winter, Mar–May Spring, Jun–Aug Summer, Sep–Nov Fall.
- **Route** is directed (**Origin-Dest**). **Cancellation_Reason**: A = Carrier, B = Weather, C = NAS, D = Security.

8. The 48-hour timeline at a glance

Your personal brief has your own hour-by-hour schedule.

Hours	What happens	Milestone
Day 1 — work on the 1-month sample		
Hours 0–3	Health check; VM or Docker decided by hour 2. Contract agreed at hour 1. ClickHouse + dummy table up. Power BI ↔ ClickHouse test. Schema audit, label and features fixed, pages wireframed.	Connection proven by hour 3
Hours 3–8	Sample in HDFS; CSV → Parquet on the sample; Tables 1 and 4; ML feature pipeline; first page layouts.	Tables 1 and 4 to Member 4 by hour 8
Hours 8–14	Sqoop (2-hour time-box), then Airflow setup. Tables 2, 3, 5, 6, 7. Baseline model. Pages 1 and 2.	All sample CSVs by hour 12
Hours 14–20	All 8 ClickHouse tables created. Sample aggregates loaded. Airflow DAG skeleton. Predictions sample. Pages 3 and 4.	

Hours	What happens	Milestone
Hours 20–24	Airflow draft DAG. ClickHouse tables validated against raw numbers. Predictions loaded. End-to-end Power BI test.	End-to-end test on the sample
Day 2 — switch to the full 48 months		
Hours 24–30	48 CSVs bulk-loaded. CSV → Parquet once on all 48 months , then aggregates. Model trained on a 10–20% sample of 2022–2023, tested on 2024. Slides 1–6.	
Hours 30–36	Run times logged. Final aggregates bulk-loaded into ClickHouse, row counts verified. ML metrics documented. Power BI refreshed on full tables.	
Hours 36–42	Airflow DAG triggered, green tree view screenshot. Q1–Q8 verification queries. ROC and feature-importance visuals. Slides 7–12.	
Hours 42–48	Master <code>README.md</code> , code cleanup, demo script. Two full rehearsals (10–12 min).	SUBMIT

9. Risks and fallbacks

Risk	Fallback
VM daemons fail	Switch to Docker Compose within the first 2 hours
Sqoop will not run	Stop after 2 hours; use <code>hdfs dfs -put</code> ; be transparent in the deck
Airflow install fails on CentOS	Python ≥ 3.9 venv, or official Airflow Docker image (<code>standalone</code>)
Power BI cannot reach ClickHouse	Test in hours 0–3. Check ODBC driver, Power BI version, port 8123 / VM network. Fallback: export tables to CSV and use Import mode
Full-data Spark runs too slow	Always read Parquet, never raw CSV; select only needed columns
ML AUC looks low	Expected with leak-free features; report honestly. Do not add leaking features
Running behind schedule	Use the cut order in section 10 below.

10. If we fall behind, cut in this order

CUT ORDER

- ML model and Power BI page 4** (not in the original project objectives).
- Airflow polish:** keep a simple DAG of `BashOperators` with one green run.
- Extra visuals** beyond the 8 questions.

Never cut: CSV→Parquet, the 7 aggregate tables, ClickHouse, and Power BI pages 1–3.